

COPD Exacerbation

Inpatient management for IM residents — bundle, oxygen, NIV, discharge

Step 1 — Triage & the oxygen target

Define the exacerbation; decide admission; set SpO₂ 88–92%

- Exacerbation = acute ↑ in dyspnea / cough / sputum beyond daily variation
- Admit for: rest dyspnea, severe baseline disease, failed outpatient Rx
- Admit for: new hypoxemia/hypercapnia, altered mentation, poor home support
- Oxygen target SpO₂ 88–92% — NOT 94–100%
 - Hyperoxia worsens hypercapnia: V/Q mismatch + Haldane effect + ↓ drive
- Get an early ABG/VBG to identify CO₂ retainers

PEARL The one number where higher is worse — keep the COPD patient at 88–92%.

Step 2 — The core bundle

Bronchodilators + steroids + criteria-based antibiotics

Element	What	Note
Bronchodilators	SABA ± SAMA, scheduled + PRN	Nebs or MDI-spacer equivalent
Steroids	Prednisone 40 mg PO × 5 d	REDUCE: oral = IV, no taper
Antibiotics	If ≥2 Anthonisen or ventilated	Cover H. flu/pneumo/Moraxella ×5 d; +Pseudomonas if risk
Oxygen	Titrate to 88–92%	Avoid hyperoxia

PEARL Sputum purulence is the strongest single trigger for antibiotics.

Step 3 — NIV: the mortality lever

BiPAP for acute hypercapnic respiratory failure

START NIV when

- $\text{pH} \leq 7.35$ with $\text{PaCO}_2 > 45$
- Severe dyspnea, accessory muscle use
- Persistent hypoxemia despite O_2

INTUBATE when

- NIV failing / pH still falling
- Cannot protect airway, $\downarrow$ consciousness
- Hemodynamic instability, apnea

PEARL NIV lowers intubation and mortality. HFNC is supportive; NIV is first-line for hypercapnic acidosis.

Find the trigger — and the mimics

- Most common trigger: viral & bacterial respiratory infection
- Also: air pollution, medication nonadherence
- Exclude PNEUMONIA — chest X-ray
- Exclude HEART FAILURE — exam, BNP
- Exclude PULMONARY EMBOLISM — any unexplained exacerbation
- Also: pneumothorax, arrhythmia

PEARL Sudden pleuritic dyspnea with no sputum change and a clear CXR → think PE, not exacerbation.

Discharge — prevent the next exacerbation

DO

- Check/optimize inhaler technique
- Step up maintenance: LAMA+LABA ($\pm$ ICS if eos $\uparrow$)
- Smoking cessation + vaccines
- Pulmonary rehab referral ≤ 4 weeks (mortality benefit)
- Action plan + early follow-up

AVOID

- Antibiotics without purulence (and not ventilated)
- Long steroid courses / tapers
- SpO₂ targets of 94–100%
- Assessing home O₂ need at acute discharge

PEARL GOLD 2026: any ≥ 1 moderate/severe exacerbation now classifies a patient as 'exacerbated.'

Take-home points

- Oxygen target 88–92% — higher is harmful
- Prednisone 40 mg PO × 5 days, no taper (REDUCE)
- Antibiotics only for ≥ 2 Anthonisen criteria or ventilation
- NIV early for hypercapnic acidosis — the biggest mortality lever
- Exclude PE; discharge with rehab, cessation, vaccines, technique